

Class Notes

Expected payoffs

Until now we considered payoff as ordered preferences. What happens if we use payoff as a measure of the utility for the player? Moreover, what happens if we allow players to randomize?

- I can compare a fixed strategy with a randomized one

Example

Consider a person that prefers option a to b and b to c. Her utilities will be $u(a) > u(b) > u(c)$. If I want to model the case where she prefers b with certainty to either a or c with probability $1/2$ I need to set $u()$ so that $u(b) > 1/2u(a) + 1/2 u(b)$

- Not all possible strategies can be modelled

Example:

Consider the following two lotteries (the first of which is, in fact, deterministic):

- Lottery 1 You receive \$2 million with certainty
- Lottery 2 You receive \$10 million with probability 0.1, \$2 million with probability 0.89, and nothing with probability 0.01.

Which do you prefer? Now consider two more lotteries:

- Lottery 3 You receive \$2 million with probability 0.11 and nothing with probability 0.89
- Lottery 4 You receive \$10 million with probability 0.1 and nothing with probability 0.9.

Most people say that they prefer lottery 1 to lottery 2 and lottery 4 to lottery 3

In the first case we have to assign a value to $u(\$10)$, $u(\$2)$ and $u(\$0)$ such that

$$u(\$2) > 0.1u(\$10) + 0.89u(\$2) + 0.01 u(\$0)$$

In the second case we have

$$0.1u(\$10) + 0.9u(\$0) > 0.11u(\$2) + 0.89u(\$0)$$

Make the system because you want an assignment that makes both true at the same time.

However,

$$u(\$2) > 0.1u(\$10) + 0.89u(\$2) + 0.01u(\$0) \quad // \text{ Move } 0.89u(\$2) \text{ to the other side in equation 1}$$

$$u(\$2) - 0.89u(\$2) > 0.1u(\$10) + 0.01u(\$0) \quad // \text{ subtract } 0.89u(\$2)$$

$$0.11u(\$2) > 0.1u(\$10) + 0.01u(\$0) \quad // \text{ add } 0.89u(\$0) \text{ both sides}$$

$$0.11u(\$2) + 0.89u(\$0) > 0.1u(\$10) + 0.01 u(\$0) + 0.89u(\$0) \quad // \text{ add } 0.89u(\$0)$$

$$0.11u(\$2) + 0.89u(\$0) > 0.1u(\$10) + 0.9u(\$0)$$

The two equations conflict!

Example: prisoner dilemma with expected payoff instead of ordinal preferences

CASE 1		Suspect 2	
		Quiet	Fink
Suspect 1	Quiet	2,2	0,3
	Fink	3,0	1,1

CASE 2		Suspect 2	
		Quiet	Fink
Suspect 1	Quiet	3,3	0,4
	Fink	4,0	1,1

With ordinal preferences CASE 1 is the same of CASE 2. However,

- in CASE 1, player 1 is indifferent between (Q,Q) for certain or (F,Q) with prob $1/2$ and (F,F) with probability $1/2$. In fact $(Q,Q) = 2$, $1/2 (F,Q) + 1/2 (F,F) = 1/2 (3) + 1/2 (1) = 2$
- in case 2, player 1 prefers (Q,Q) for certain to F,Q) with prob $1/2$ and (F,F) with probability $1/2$. In fact $(Q,Q) = 3$, $1/2 (F,Q) + 1/2 (F,F) = 1/2 (4) + 1/2 (1) = 2.5$

Mixed strategy

Let's consider the case in which instead of choosing a strategy a player can randomize following a probability distribution.

Example: Matching Pennies. An analysis of the game Matching Pennies illustrates the idea of a stochastic steady state. The discussion focuses on the outcomes of this game, rather than payoffs that represent the players' preferences, as before.

		Player 2	
		Head	Tail
Player 1	Head	\$1, -\$1	-\$1, \$1
	Tail	-\$1, \$1	\$1, -\$1

This game has no Nash equilibrium, however, the game has a stochastic steady state in which each player chooses each of her actions with probability $1/2$.

To establish this result, I need to argue that if player 2 chooses each of her actions with probability $1/2$, then player 1 optimally chooses each of her actions with probability $1/2$, and vice versa.

Suppose that player 2 chooses each of her actions with probability $1/2$. If player 1 chooses Head with probability p and Tail with probability $1 - p$ then each outcome (Head, Head) and (Head, Tail) occurs with probability $1/2p$, and each outcome (Tail, Head) and (Tail, Tail) occurs with probability $1/2(1 - p)$. Thus player 1 gains \$1 with probability $1/2p + 1/2(1 - p)$, which is equal to $1/2$, and loses \$1 with probability $1/2$. In particular, the probability distribution over outcomes is independent of p ! Thus, every value of p is optimal. In particular, player 1 can do no better than choose Head with probability $1/2$ and Tail with probability $1/2$. A similar analysis shows that player 2 optimally chooses

each action with probability 1/2 when player 1 does so. We conclude that the game has a stochastic steady state in which each player chooses each action with probability 1/2.

The game has no other steady state. In fact, if player 2 chooses Head with probability less than 1/2, (in the example below I use 0.3) then the uniquely best policy for player 1 is to choose Tail with certainty. A similar argument shows that if player 2 chooses Head with probability greater than 1/2, the uniquely best policy for player 1 is to choose Head with certainty.

Maximize payoff for p_1 setting the appropriate $(p, 1-p)$

		p_2	
		H	T
p_1	H	1, -1	1, -1
	T	-1, 1	-1, 1

0.3 0.7

$\max_{p \in [0,1]} P_{win} - P_{lose}$

$$P_{win} = 0.3p + 0.7(1-p)$$

$$P_{lose} = -0.3p - 0.7(1-p)$$

$$\max_{p \in [0,1]} (P_{win} - P_{lose}) = \max_{p \in [0,1]} (0.3p + 0.7(1-p) + 0.3p + 0.7(1-p))$$

$$= \max_{p \in [0,1]} (0.6p - 1.4p + 1.4) = \max_{p \in [0,1]} \left(-p + \frac{1.4}{0.8} \right)$$

$\rightarrow p = 0 \rightarrow \text{chose always T}$

Two observations:

- The steady state is a Mixed strategy Nash equilibrium
- Mixed strategy Nash equilibrium always exists

Find Mixed strategy Nash equilibrium

To find a mixed strategy Nash equilibrium we need to convince players to play a randomized strategy instead of a fixed one. To this end, player 1 has to make her choices of probability distribution so that player 2 is indifferent between any fixed strategy and thus she is willing to randomize. The same hold vice versa.

Let's take again the matching penny example

		Player 2	
		Head	Tail
Player 1	Head	1, -1	-1, 1
	Tail	-1, 1	1, -1

We need to find a probability distribution $(p, 1-p)$ for player 2 actions that makes player 1 indifferent. This means that $u_1(H) = u_1(T)$ where $u_1(H) = 1 \times p + -1(1-p)$ and $u_1(T) = -1 \times p + 1(1-p)$. The equation leads to $p = \frac{1}{2}$. If we repeat the same reasoning to make player 1 to fix a probability distribution $(q, 1-q)$ for her strategy to make player 2 indifferent we will find that $q = \frac{1}{2}$. Consequently, the strategy $p_1(H, T) = (\frac{1}{2}, \frac{1}{2})$ and $p_2(H, T) = (\frac{1}{2}, \frac{1}{2})$ is a mixed strategy Nash equilibrium.

Example with Back or Stravinsky

		Player 2	
		Back	Stravinsky
Player 1	Back	2,1	0,0
	Stravinsky	0,0	1,2

Again, we need to find a probability distribution $(p, 1-p)$ for player 2 actions that makes player 1 indifferent. This means that $u_1(B) = u_1(S)$ where $u_1(B) = 2 \times p$ and $u_1(S) = 1(1 - p)$. The equation leads to $p = \frac{2}{3}$. Similarly, we compute the probabilities for player 1 to make player 2 indifferent. the strategy $p_1(B, S) = (\frac{2}{3}, \frac{1}{3})$ and $p_2(B, S) = (\frac{1}{3}, \frac{2}{3})$ is a mixed strategy Nash equilibrium.

The following table computes the probability of each possible outcome

		Player 2	
		Back	Stravinsky
Player 1	Back	$\frac{2}{3} \times \frac{1}{3}$	$\frac{2}{3} \times \frac{2}{3}$
	Stravinsky	$\frac{1}{3} \times \frac{1}{3}$	$\frac{1}{3} \times \frac{2}{3}$

The expected payoff of player 1 is: $2 \left(\frac{2}{3} \times \frac{1}{3} \right) + 1 \left(\frac{1}{3} \times \frac{2}{3} \right) = \frac{4}{9} + \frac{2}{9} = \frac{6}{9} = \frac{2}{3}$

The same happens to player 2.

Consider the game of kicking penalty

		Goalkeeper	
		Left	Right
kicker	Left	0, 1	1, 0
	Right	1, 0	0, 1

This game has a mixed strategy equilibrium when both randomize with probability 0.5

Consider a kicker that sometimes fails when kicks right. Should he prefer kicking left? How does the game change?

		Goalkeeper	
		Left	Right
kicker	Left	0, 1	1, 0
	Right	.75, .25	0, 1

Computing the Mixed equilibrium, we find that GK has a strategy $u_{GK}(L) = u_{GK}(R) \rightarrow p = 0.75p \rightarrow p = \frac{1}{1.75} = \frac{4}{7}$. For K the strategy $u_K(L) = u_K(R) \rightarrow p + 0.25(1 - p) = 1 - p \rightarrow p = \frac{0.75}{1.75} = \frac{3}{7}$

The mixed equilibrium is $p_K(L, R) = (\frac{4}{7}, \frac{3}{7})$ and $p_{GK}(L, R) = (\frac{3}{7}, \frac{4}{7})$

Notice that the kicker should prefer his weaker side in response of the fact that the goalkeeper tend to defend more the stronger side. Computing the expected payoffs, we see that the probability of winning for the goalkeeper is higher $\frac{4}{7}$ vs $\frac{3}{7}$. This is in line with the weakness of the kicker.

A real example [1] "Professionals play minimax" can be seen here:
<https://www.youtube.com/watch?v=Vka32gFzPiY>

Example (Traffic light) The game we consider is when two players drive up to the same intersection at the same time. If both attempt to cross, the result is a fatal traffic accident. The game can be modeled by a payoff matrix where crossing successfully has a payoff of 1, not crossing pays 0, while an accident costs -100 .

		Player 2	
		Cross	Wait
Player 1	Cross	-100, -100	1, 0
	Wait	0, 1	0, 0

This game has three Nash equilibria: two correspond to letting only one car cross, the third is a mixed equilibrium where both players cross with an extremely small probability $q = 1/101$, and with q^2 probability they crash.

The first two equilibria have a payoff of 1. The last one is more fair, but has low expected payoff (≈ 0.0001), and also has a positive chance of a car crash.

In a Nash equilibrium, players choose their strategies independently. In contrast, in a correlated equilibrium a coordinator can choose strategies for both players; however, the chosen strategies have to be stable: we require that each player find it in his or her interest to follow the recommended strategy.

For example, in a correlated equilibrium the coordinator can randomly let one of the two players cross with any probability. The player who is told to stop has 0 payoff, but he knows that attempting to cross will cause a traffic accident.

Reference

- Palacios-Huerta, Ignacio. "Professionals play minimax." *The Review of Economic Studies* 70.2 (2003): 395-415.